Space Invaders

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Introduction

This project involves the design and development of a 2D arcade-style game inspired by the classic Space Invaders. The game is built using Python and the Pygame library, incorporating interactive gameplay, enemy stages, sound effects, and player progression. The aim is to enhance programming, logic, and design skills through game development.

Objectives

- To design an engaging and interactive 2D shooting game using Pygame.
- To implement multiple stages with increasing difficulty levels.
- To integrate sound effects, score tracking, and player health/lives.
- To gain practical experience in object-oriented programming and game loops.

Literature Review

Author(s)	Title	Year	Strength	Limitation
Smith et al.	*Developing Games with Pygame*	2020	Clear use of Pygame features	Lacks advanced enemy behavior
Johnson & Lee	*Retro Game Design Fundamentals*	2019	Emphasizes level design	Limited discussion on animations
Kim & Park	*Gamification in Education*	2021	Shows educational value of games	Not focused on technical details

Methodology

1. Used Pygame to create a windowed interface and render assets.
2. Designed sprite classes for the player, bullets, and enemies.
3. Implemented stage-based logic with increasing enemy speed and type.
4. Added sound integration for shooting, explosions, and background music.
5. Applied structured OOP for modular and reusable code.

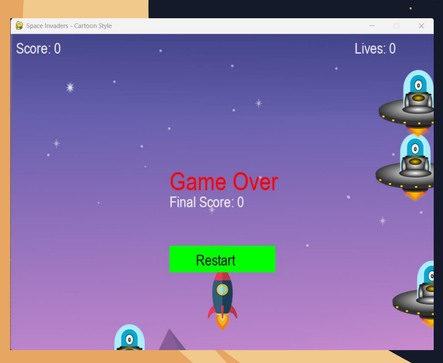
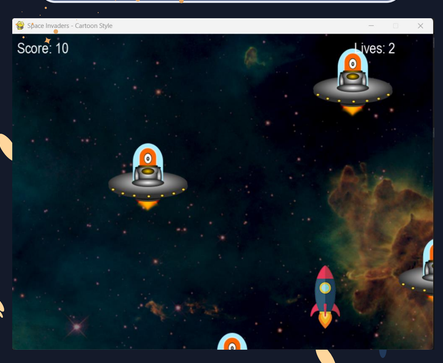
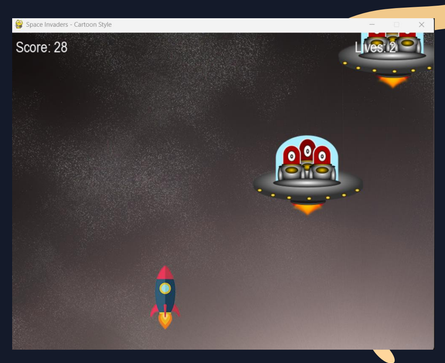
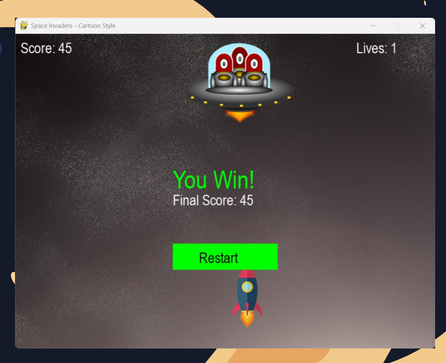
Results

1. Successfully developed a 3-stage game where difficulty increases at each stage.
2. Implemented score tracking, lives system, and restart functionality.
3. Added pop-up animations for level transitions and a final win screen.
4. Integrated explosion and shooting sound effects for player feedback.

Limitations

1. The game is single-player only.
2. Enemy movement is simple and lacks adaptive behavior.
3. No saving/loading of high scores.
4. Graphics are basic and cartoon-styled for simplicity.

Conclusion

This project demonstrates a strong understanding of game development principles using Python and Pygame. It successfully implements core gameplay mechanics, stage logic, and interactive feedback. Future improvements can include multiplayer features, AI-based enemies, and advanced UI/UX enhancements.

Outcomes